

BPHO ROUND 0 · QUESTION PAPER

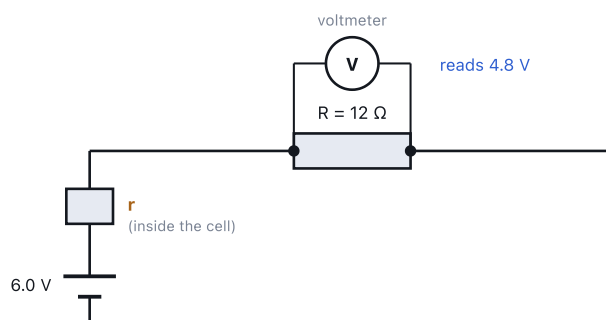
Drill bank section 4

25 questions · 60 minutes · single answer, 5 options · **no calculator** · no negative marking*Original questions written for this course, in the Round 0 format. Not BPhO material.***Before you start**

- Answer every question. There is no negative marking, so a blank is a wasted mark.
- No calculator. Every answer here is reachable by hand, and the numbers are chosen to divide cleanly.
- Work on paper, then mark yourself against the markscheme PDF.
- The 2025 UK qualifying line was 11 out of 25, so about 44% is the target to beat.

Q1 · A cell with internal resistance: how the terminal p.d. changes with the load H

The figure shows a cell of emf 6.0 V whose internal resistance r is inside the cell, connected in series with a $12\ \Omega$ resistor. The voltmeter is connected across the $12\ \Omega$ resistor and reads 4.8 V.



The $12\ \Omega$ resistor is now replaced by a $6.0\ \Omega$ resistor. What does the voltmeter read?

- | | |
|----------------|----------------|
| A 2.4 V | B 3.0 V |
| C 4.0 V | D 4.8 V |
| E 6.0 V | |

Q2 · Dimensional analysis: finding the combination of G , M and r that has the dimensions of a time A

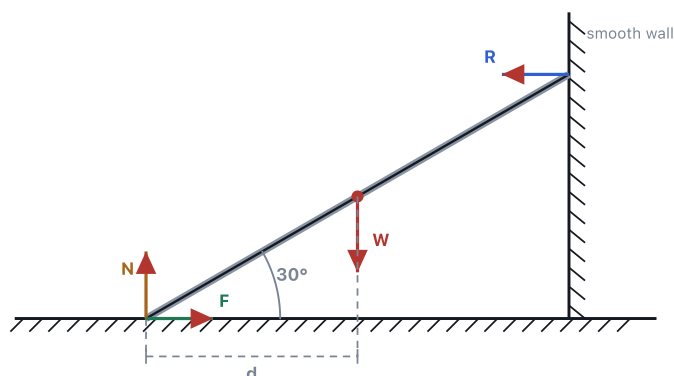
A satellite moves in a circular orbit of radius r about a planet of mass M . The only quantities available to describe the motion are r , M , and the gravitational constant G , whose units are $\text{N m}^2 \text{kg}^{-2}$.

Which of the following has the dimensions of a time?

- | | |
|--------------------------------|------------------------------|
| A $2\pi\sqrt{(r^3/GM)}$ | B $2\pi\sqrt{(GM/r)}$ |
| C $2\pi\sqrt{(GM/r^3)}$ | D $2\pi r^3/(GM)$ |
| E $2\pi GM/r^2$ | |

Q3 · A uniform ladder against a smooth wall: the smallest coefficient of friction at the foot **C**

The figure shows a uniform ladder of weight W and length L resting with its foot on rough horizontal ground and its top against a smooth vertical wall. The ladder makes an angle of 30° with the ground.



The wall exerts only a horizontal reaction. Which expression gives the smallest coefficient of friction between the foot and the ground that can hold the ladder in equilibrium?

- | | |
|-----------------------|-----------------------|
| A $1/2$ | B $\sqrt{3}/2$ |
| C $\sqrt{3}$ | D $1/\sqrt{3}$ |
| E $2/\sqrt{3}$ | |

Q4 · A uniform wire bent into an equilateral triangle: the resistance between two vertices **H**

A uniform wire of total resistance $12\ \Omega$ is bent into an equilateral triangle. A battery is connected between two of the vertices. What is the resistance of the network between those two vertices?

- | | |
|------------------------|------------------------|
| A $1.3\ \Omega$ | B $4.0\ \Omega$ |
| C $6.0\ \Omega$ | D $8.0\ \Omega$ |
| E $2.7\ \Omega$ | |

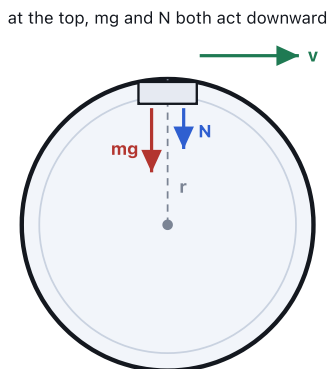
Q5 · Order-of-magnitude estimation: the number of heartbeats in a lifetime **A**

A human heart beats about 70 times a minute at rest. Taking a lifetime to be 80 years, which of the following is the order of magnitude of the number of heartbeats in a lifetime?

- | | |
|-------------------------------|-------------------------------|
| A 3.7×10^7 | B 3.0×10^8 |
| C 2.9×10^9 | D 1.3×10^{10} |
| E 1.8×10^{11} | |

Q8 · The top of a vertical circular loop: the smallest speed that keeps contact **D**

The figure shows a car at the top of a vertical circular loop of radius r . At that point the weight and the normal reaction from the track both act downward, towards the centre of the circle.



so $mg + N = mv^2/r$, and the smallest possible N is zero

What is the smallest speed the car can have at the top and still stay in contact with the track?

- | | |
|-----------------------|-----------------------|
| A $\sqrt{2gr}$ | B $\sqrt{g/r}$ |
| C gr | D $\sqrt{gr}$ |
| E $2\sqrt{gr}$ | |

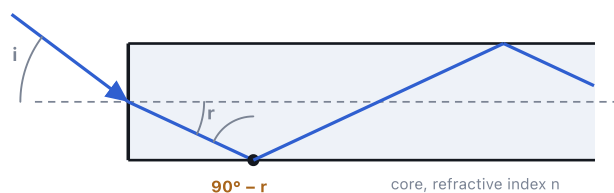
Q9 · The speed of a wave on a stretched string when the tension is increased **F**

A transverse wave travels along a stretched string at 40 m/s. The tension in the string is increased by 21%, the string being kept at the same length. What is the new wave speed?

- | | |
|-----------------|-----------------|
| A 36 m/s | B 41 m/s |
| C 48 m/s | D 58 m/s |
| E 44 m/s | |

Q10 · An optical fibre: the condition for a ray to be guided along the core **G**

The figure shows a ray entering the flat end face of an optical fibre from air at an angle i to the axis. Inside the core, of refractive index n , the ray travels at an angle r to the axis and then meets the side wall, where the normal is horizontal.

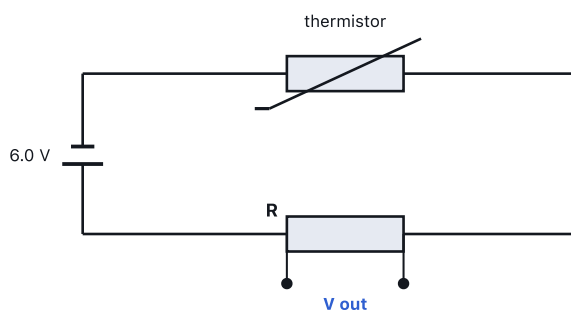


Which expression gives the largest value of $\sin i$ for which the ray is totally internally reflected at the side wall?

- | | |
|-----------------------------|------------------|
| A $\sqrt{(n^2 - 1)}$ | B $n - 1$ |
| C $\sqrt{(n^2 + 1)}$ | D $1/n$ |
| E $(n^2 - 1)/n$ | |

Q11 · A thermistor in the top arm of a potential divider: the output across the lower arm **H**

The figure shows a potential divider across a 6.0 V supply with a thermistor in the upper arm and a fixed resistor of 2.0 k Ω in the lower arm. The output is taken across the fixed resistor.



When the thermistor is warm its resistance has fallen to 1.0 k Ω . What is the output voltage then?

- | | |
|----------------|----------------|
| A 1.5 V | B 2.0 V |
| C 3.0 V | D 4.0 V |
| E 4.5 V | |

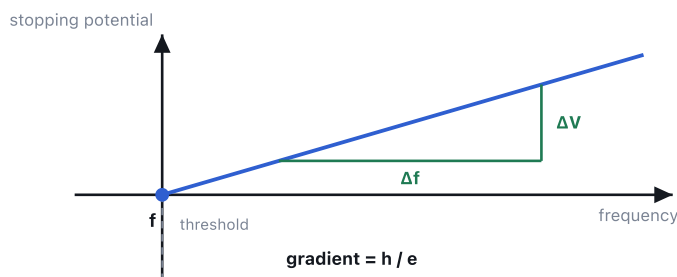
Q12 · The density of nuclear matter and the dependence of nuclear radius on mass number **K**

The radius of a nucleus of mass number **A** is given by $R = r_0 A^{1/3}$, where **r0** is a constant. What is the ratio of the density of nuclear matter in a nucleus of mass number **8A** to that in a nucleus of mass number **A**?

- A** $1/8$ **B** 1
C 2 **D** 4
E 8

Q13 · The gradient of a stopping-potential against frequency graph, and what it measures **L**

The figure shows the stopping potential for a metal plotted against the frequency of the incident light. The graph is a straight line that crosses the frequency axis at the threshold frequency, and the gradient triangle marked on the line gives a gradient of $4.0 \times 10^{-15} \text{ V s}$.



Taking the electronic charge to be $1.6 \times 10^{-19} \text{ C}$, what is the value of Planck's constant obtained from this graph?

- A** $6.4 \times 10^{-35} \text{ J s}$ **B** $2.5 \times 10^4 \text{ J s}$
C $4.0 \times 10^{-15} \text{ J s}$ **D** $6.4 \times 10^{-16} \text{ J s}$
E $6.4 \times 10^{-34} \text{ J s}$

Q14 · Finding the base units of the viscosity in Stokes' law **A**

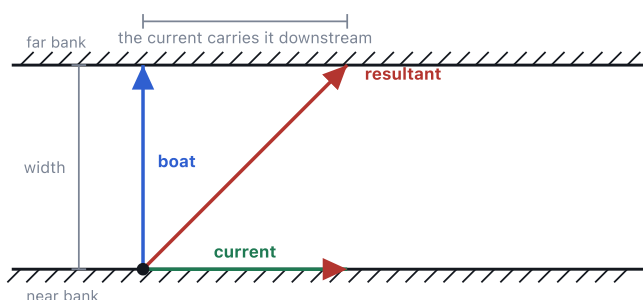
The drag force on a small sphere moving slowly through a fluid is given by $F = 6\pi\eta rv$, where **r** is the radius of the sphere, **v** is its speed, and **η** is the viscosity of the fluid.

Which of the following is the base-unit form of the viscosity?

- A** kg m s^{-1} **B** $\text{kg m}^{-2} \text{ s}^{-1}$
C $\text{kg m}^{-1} \text{ s}^{-1}$ **D** $\text{kg m}^{-1} \text{ s}^{-2}$
E kg s^{-1}

Q15 · Crossing a river: the time taken depends only on the component of velocity across the river **B**

The figure shows a boat leaving the near bank pointed straight across a river, with the current flowing along the bank. The river is 120 m wide and the boat's speed relative to the water is 3.0 m/s. The current is 2.0 m/s.

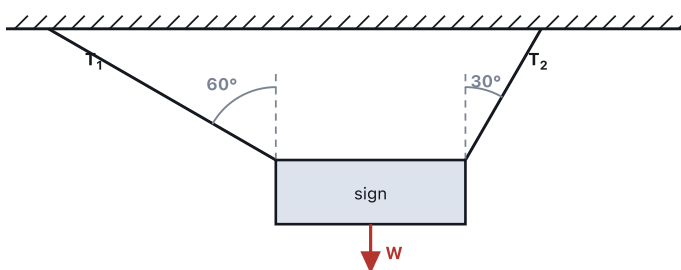


How long does the crossing take?

- | | |
|---------------|---------------|
| A 24 s | B 33 s |
| C 48 s | D 40 s |
| E 60 s | |

Q16 · A sign hanging from two wires at unequal angles: which wire carries more of the load **C**

The figure shows a sign of weight W hanging from two wires attached to a ceiling. The left-hand wire makes 60° with the vertical and the right-hand one makes 30° with the vertical.



Which expression gives the tension in the right-hand wire?

- | | |
|--------------------------|------------------------|
| A $W / 2$ | B $W\sqrt{3}/2$ |
| C W | D $2W$ |
| E $2W / \sqrt{3}$ | |

Q17 · A conical pendulum: the period in terms of the string length and the cone angle **D**

A small bob is attached to a light string of length **L** and moves in a horizontal circle, the string making a constant angle **θ** with the vertical throughout the motion. Which expression gives the period of the motion?

- A** $2\pi\sqrt{(L \cos\theta / g)}$
B $2\pi\sqrt{(L / g)}$
C $2\pi\sqrt{(L / (g \cos\theta))}$
D $2\pi\sqrt{(g / (L \cos\theta))}$
E $2\pi\sqrt{(L \sin\theta / g)}$

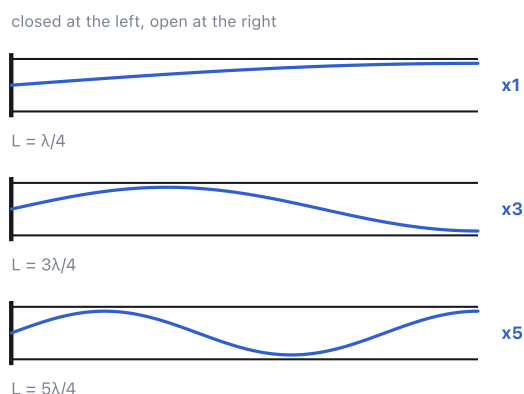
Q18 · Two wires of the same material and length but different diameters, under the same load **E**

Two wires are made of the same material and have the same length. The second wire has twice the diameter of the first. The same load is hung on each. What is the ratio of the extension of the second wire to that of the first?

- A** $1/2$
B 1
C 2
D 4
E $1/4$

Q19 · A pipe closed at one end: finding its length from two successive resonances **F**

The figure shows the three lowest standing-wave patterns in a pipe closed at one end and open at the other.



The pipe resonates at 400 Hz and at the next higher resonance at 1200 Hz. Taking the speed of sound in air to be 340 m/s, what is the length of the pipe?

- A** 0.11 m
B 0.43 m
C 0.21 m
D 0.85 m
E 1.7 m

Q20 · The critical angle at a boundary with air for a material of refractive index root two **G**

A transparent material has refractive index $\sqrt{2}$. What is the largest angle of incidence for which light travelling in the material can emerge into the air?

- A** 30° **B** 35.3°
C 54.7° **D** 45°
E 90°

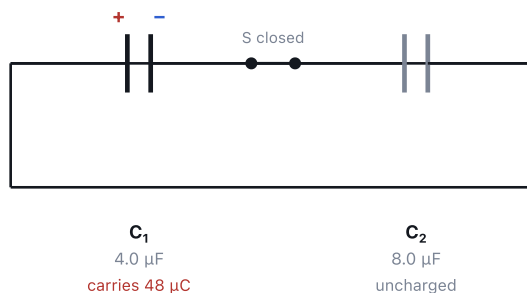
Q21 · A series resistor feeding a parallel pair: the current drawn from the battery **H**

A battery of emf 6.0 V and negligible internal resistance is connected to a $2.0\ \Omega$ resistor in series with a parallel combination of a $6.0\ \Omega$ and a $3.0\ \Omega$ resistor. What is the current drawn from the battery?

- A** 0.75 A **B** 1.0 A
C 2.0 A **D** 3.0 A
E 1.5 A

Q22 · Charge sharing between a charged and an uncharged capacitor: the common potential difference **I**

The figure shows a $4.0\ \mu\text{F}$ capacitor carrying a charge of $48\ \mu\text{C}$, connected through a closed switch to an uncharged $8.0\ \mu\text{F}$ capacitor.



What is the potential difference across the two capacitors once the charge has settled?

- A** 2.0 V **B** 4.0 V
C 6.0 V **D** 8.0 V
E 12 V

Q23 · Mixing two masses of water: the final temperature when no heat is lost **J**

In an insulated container, 0.40 kg of water at 90°C is mixed with 0.10 kg of water at 10°C . What is the final temperature of the mixture?

- A** 26°C **B** 37°C
C 50°C **D** 74°C
E 82°C

Q24 · Radioactive decay: the fraction of a sample remaining after three half-lives K

A radioactive isotope has a half-life of 8.0 days. What fraction of the original number of nuclei remains after 24 days?

- | | |
|-----------------|----------------|
| A $1/24$ | B $1/3$ |
| C $1/8$ | D $1/2$ |
| E $3/8$ | |
-

Q25 · An electron and a proton with the same de Broglie wavelength: the ratio of their kinetic energies L

An electron and a proton have the same de Broglie wavelength. Taking the mass of the proton to be 1836 times the mass of the electron, what is the ratio of the electron's kinetic energy to the proton's?

- | | |
|----------------------------|-------------|
| A 1836 | B 1 |
| C $1/1836$ | D 43 |
| E 3.4×10^6 | |